MIL-T-81783C(AS)
11 July 1977
Superseding
MIL-T-81783B(AS)
6 December 1974

MILITARY SPECIFICATION
TEST STATION, CONTROLS-DISPLAYS-DOPPLER FILTER OJ-214( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Controls-Displays-Doppler Filter Test Station (C&D TS) OJ-214( )/AWM-23(V).

Classification. The C&D TS consists of a 2-bay station as follows (see 6.4):

Bay No. 1: Console, Test, Control-Display-Doppler Filter OJ-251( )/AWM-23(V)

Nomenclature	Reference Paragraph
(1) Panel, Power Distribution SB-3695( )/AWM-23(V)	3.6.1.1
(2) Digital Multimeter	3.6.1.2
(3) Counter/Timer	3.6.1.3
(4) Switching Unit, Fault Isolation SA-1894( )/AWM-23(V)	3.6.1.4
(5) Mounting MT-4500( )/AWM-23(V)	3.6.1.5
(6) Mounting MT-4499( )/AWM-23(V)	3.6.1.6
(7) Drawer, Storage*	3.6.1.7
(8) Extender Kit, C&D TS	3.6.1.7.1

*Two required

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, N.J. 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-4935

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

Nomenclature	Reference Paragraph
(9) Adapter, Test-Hand Control	3.6.1.8

b. Bay No. 2: Console, Test, Control-Display-Doppler Filter OJ-252( )/AWM-23(V)

(1) Oscilloscope	3.6.2.1
(2) Panel, Test, Electrical C-9178( )/AwM-23(V)	3.6.2.2
(3) Panel, Coolant Supply MX-9306( )/AWM-23(V)	3.6.2.3
(4) Mounting MT-4501( )/AWM-23(V)	3.6.2.4
(5) Panel, Blank	3.6.2.5
(6) Power Supply PP-6849( )/AWM-23(V)	3.6.2.6
(7) Power Supply, DC	3.6.2.7

1.3 Associated equipment. The C&D TS shall operate with Kit, Tool, Special TK-222( )/AWM-23(V).

2 APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in the AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the C&D TS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

Hughes Aircraft Company

3282205	Acceptance Test Procedure, Test Station, Controls-Displays-Doppler Filter

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any other document.

b. This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).

c. Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

3.2 Design and construction. Applicable except as follows.

3.2.1 Design features. Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. The weight of the C&D TS shall not exceed 1600 pounds. Maximum dimensions shall be 48 inches by 54 inches by 75 inches with all drawers and retractable mounting fixtures in the retracted position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

3.2.1 .9.1 Storage space. All items which are not permanently attached to the C&D TS shall be stored at the C&D TS. Two storage drawers (see 3.6.1.7), 10 inches by 19 inches by 20 inches, are contained in bay no. 1. These drawers shall hold card extenders, extender cables, test leads, card extractors, and tools.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary power. The total secondary power shall be limited to 1600 watts maximum, with any combination of power form loading within the load capacity limits for each power form listed in Table I.

TABLE I
SECONDARY POWER FORMS, SEMI REGULATED

Item	Power Form (Vdc)	Load Capacity (Amperes)
1	+9.7 $^{+1.6}_{-1.8}$	45.0
2	+17.0 $^{+2.8}_{-1.8}$	33.0
3	+31.0 $^{+5.1}_{-3.4}$	23.0
4	+57.0 $^{+5.7}_{-4.5}$	6.0
5	+73.0 $^{+7.3}_{-9}$	0.6
6	+120.0 $^{+12}_{-10}$	0.3
7	+320.0 $^{+32}_{-30}$	0.12
8	-17.0 $^{-2.8}_{+1.8}$	18.0
9	-31.0 $^{-5.1}_{+3.4}$	16.0
10	-57.0 $^{-5.7}_{+4.5}$	4.0
11	-162.0 $^{-16}_{+19.5}$	0.16

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

3.2.1.11.1 Secondary power semi-regulated. Secondary electrical power forms shall be provided by power supplies which are part of the C&D TS. Power forms, from 10 percent to full load, and for normal values of line voltage variations, shall be as listed in Table I.

3.2.1.11.2 Secondary power, regulated. Secondary regulated electrical power forms shall be provided by power supplies which are part of the C&D TS. Regulation from no load to full load, and for normal values of line voltage variations, shall not exceed 2 percent. Ripple voltage for dc power forms shall not exceed 150 millivolts (mV) peak-to-peak (p-p). The power forms listed in Table II shall be provided.

3.2.1.12 Interface. The C&D TS shall provide the following interfaces.

3.2.1.12.1 Clean air interface. The C&D TS shall interface with the clean air through ducts which enter the top rear of each test console of the C&D TS.

3.2.1 .12.2 Electrical power interface, The C&D TS shall interface with electrical power for the convenience outlets at one end of the convenience outlet strip mounted on the top of the front face of the C&D TS. The C&D TS shall interface with all other electrical power through power lines entering the top of bay number 1.

TABLE II
SECONDARY POWER FORMS, REGULATED

Item	Power Form	Load Capacity (Amperes)*
1	+5Vdc	20.0
2	+12Vdc	1.6
3	+25Vdc	3.0
4	+50Vdc	3.5
5	-12Vdc	0.6
6	-25Vdc	0.5

* The regulated power voltage currents are part of the load capacity of 3.2.1.11.1

3.2.2 Utilities. Applicable except as follows.

3.2.2.1 Electrical power. Applicable. Electrical power required for operation of the C&D TS shall be as listed in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

TABLE III
C&D TS ELECTRICAL POWER

Item	Input Power Form	Maximum Load
1	+29±4 Vdc	0.28 Kilowatt (kW)
2	115 Vac, 400 Hertz (Hz), 3-phase, 4-wire, Type III	0.5 kilovolt Amperes (kVA)
3	115 Vac, 60Hz, 1-phase, Type I	1.75 kVA
4	115 Vac, 60Hz, 3-phase, 3-wire, ungrounded, Type I	1.9 kVA

3.2.2.2 Clean air. Applicable. The C&D TS requirement for forced-air shall not exceed 25 pounds per minute.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3,2.3 Service conditions.

3.2.3.1 Electrical. Applicable.

3.2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Applicable, with the following exceptions.

3.2.4.1 Mean time to repair. Applicable. The C&D TS shall provide capability to isolate faults to:

a. The instrument level for commercial test equipment.

b. SRA level for peculiar equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

3.2.4.2 Test points. Applicable. Test points shall be provided to facilitate fault isolation to the SRA level, fault isolation within an SRA, alignments or adjustments, calibration, and signal monitoring. Location and accessibility of test points shall be dictated by considerations of ease of maintenance and repair.

3.2.4.3 Shop replaceable assemblies (SRAs). Applicable.

3.2.4.4 Self test capability. The C&D TS shall provide capability to establish its functional readiness. Time to perform self test shall not exceed 30 minutes.

3.2.5 Reliability. The mean time between failures shall be 120 hours, minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Applicable except as follows:

a. MIL-STD-461, tests CE03 and RS02-1 and the limits therein shall apply.

b. MIL-STD-461, test CE04 shall apply only when power remote contol leads external to test station are used.

c. MIL-STD-461, tests CE01, CE02, RS01,RS02-2, and RS04 shall not apply.

3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection. Applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.7.3.1 Secondary voltage protection. Occurrence of an out-of-tolerance secondary voltage condition that could damage either an item under test or the test station shall result in automatic shut down of the secondary power forms. The out-of-tolerance condition leading to shutdown shall be identified by a projection readout which displays the power form shut down and indicator lamps to show whether the power form is regulated or unregulated. Operator action shall be required to reinstate test station operation after the occurrence of automatic shutdown.

3.2.7.3.2 Emergency shutdown switch. An emergency shutdown switch shall be provided for use by the operator. When the emergency shutdown switch is activated, the load side of all C&D TS primary circuit breakers shall be disconnected from their respective loads. The switch utilized for this application

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

shall be of the pushbutton-activated, momentary type, with the pushbutton finished in red. The front panel area around the pushbutton shall be marked with alternating black and orange-yellow stripes. The switch shall be centrally located on the C&D TS in an area which is easily accessible to the operator during normal operation.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching and coolant control. The C&D TS shall provide the necessary switching to control utility electrical power forms and shall provide orifices and flow switches for control of utility supplied clean air.

3.2.10 Displays. The C&D TS displays shall provide an indication of C&D TS operational and test status.

3.3 Performance.

3.3*1 General. The C&D TS shall provide the capability to test and maintain the equipment listed in Table IV.

3.3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warm-up time. Applicable.

3.4 Detailed functional requirements.

3.4.1 Test capabilities. The C&D TS shall provide and control the stimuli, the power forms, and the loads required to perform functional and diagnostic testing of the weapon replaceable assemblies (WRAs) listed in Table IV. Test capabilities provided by the C&D TS shall be as follows.

3.4.1.1 Filter-Multiplexer F-1331( )/AWG-9. The C&D TS shall provide the capability to perform the following tests on the filter-multiplexer assembly:

a. Built-in-test (BIT) power fault. With power applied, measure BIT power fault signal while monitoring the regulated voltage levels on the test access provided.

b. Sub band amplifier response. Apply proper enabling signals along with a radio frequency (RF) signal and measure intermediate frequency (IF) amplitude at the test access lines, Adjust RF signal frequency to various points along the amplifier pass band and repeat measurement.

c. Local oscillator frequencies. Apply enabling signals and RF input. Mix RF input and sub band IF together and count the difference or local oscillator (LO) frequency. Change selection logic inputs to check other LO frequencies.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

TABLE IV
WRAs MAINTAINED AT THE C&D TEST STATION

Item	WRA Nomenclature (see 6.4)
1	Filter-Multiplexer F-1331()/AWG-9
2	Control, Sensor C-9035()/AWG-9
3	Control, Computer C-9036()/AWG-9
4	Display, Detail Data IP-1111()/AWG-9, IP-1173()/AWG-9 and IP-1185()/AWG-9
5	Control, Sensor C-9037()/AWG-9
6	Display, Tactical IP-1110()/AWG-9
7	Recorder, Mission Data KF-21() and KF-22()

d. BIT Automatic Gain Control (AGC). Apply all enabling signals and various levels of RF signals and measure the dc voltage on the BIT AGC line to an accuracy of + 1.0 percent.

e. AGC time constants. Apply all enabling signals and a square wave modulated RF signal. Using an oscilloscope connected to the BIT AGC line measure the various time constants to an accuracy of + 5 percent.

f. Phase identification and phase timing. Apply Enable FMR signal and with the counter, measure relative timing of Phase Identification, A, B, C, and Phase Timing Signals.

g. Threshold control. Apply Mode Control logic and measure threshold levels at test access provided.

h. Filter ringing test. Apply all enabling signals. Using a voltage controlled oscillator, sweep through the entire filter bank and observe the output video on an oscilloscope to check for proper ringing conditions,

3.4.1.2 Control, Sensor C-9035()/AWG-9. The C&D TS shall provide the capability for performing the following tests on the sensor control assembly:

a. Verify four position RDR EL BARS rotary switch operation.

b. Verify four position RDR AZ SCAN rotary switch operation.

c. Verify RDR AZ CNTR potentiometer operation.

d. Verify RDR EL CNTR potentiometer operation.

e. Verify four position IR EL BARS rotary switch operation.

Source: https://assist.dla.mil -- Downloaded: 202640-23T06:09Z
Check the source to verify that this is the current version before use.

f. Verify four position IR AZ SCAN rotary switch operation.

g. Verify IR/TV AZ CNTR potentiometer operation.

h. Verify IR/TV EL CNTR potentiometer operation.

i. Verify three position SLAVE toggle switch operation.

j. Verify two position STAB toggle switch operation.

k. Verify three position VSL toggle switch operation (RADAR).

l. Verify three position TRACK toggle switch operation.

m. Verify FOCUS potentiometer operation.

n. Verify RDR/TV TRIM (AZ) potentiometer operation.

o. Verify RDR/TV TRIM (EL) potentiometer operation.

p. Verify regulating function of voltage regulators.

3.4.1.3 Control, Computer C-9036()/AWG-9. The C&D TS shall provide the capability for performing the following tests on the computer control assembly:

a. Verify digital address group operation.

(1) Data Entry Keyboard (12 pushbuttons)

(2) CLR pushbutton

(3) Enter pushbutton

(4) Master control function switches (10 pushbuttons - arrow)

b. Verify TNG pushbutton operation.

c* Verify PRGM RESTRT pushbutton operation.

d. Verify six position master controls array rotary switch and six sided drum indicator operation.

e. Verify operation of standard serial interface.

3.4.1.4 Display, Detail Data IP-1111()/AWG-9, IP-1173()/AWG-9 and IP-1185()/AWG-9. The C&D TS shall provide the capability for performing the following tests on the detail data display (DDD) assembly:

a. Display and check presence, distortion and positioning, accuracy of test patterns.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

b. Verify WCS MODE group pushbutton switch operation.

c. Verify DISPLAY push button switch operations.

d. Verify WCS annunciator readouts.

e. Verify three position TGTS toggle switch operation.

f. Verify three position target ASPECT toggle switch operation.

g. Verify three position MLC toggle switch operation.

h. Verify twelve position XMTR CHAN digiswitch operation.

i. Verify six position MSL CHAN digiswitch operation.

j. Verify DDD potentiometer control operation.

(1) ERASE

(2) IR GAIN

(3) PULSE VIDEO

(4) IR AUDIO VOL

(5) IR AUDIO THRLD

k. Verify five position JAM/JET rotary switch operation.

l. Verify BRIGHT mechanical filter operation.

m. Verify jewel light operation.

(1) ANT TRK

(2) RDROT

(3) JAT

(4) IROT

n. Verify Range Scale Select pushbutton switch operation.

o. Verify RANGE one position readout.

p. Verify operation of ACM THRLD potentiometer and switch.

q. Verify operation of CCM Modes pushbutton switches.

r. Verify VGS pushbutton switch operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

s. Verify RDR-IR/TV/EC ELEV meter operation.

t. Verify two position PARAMP toggle switch operation.

u. Verify two position AGC toggle switch operation.

v. Verify three position PD THRLD CLEAR potentiometer and switch operation.

w. Verify PD THRLD CLUTTER potentiometer and switch operation.

x. Verify operation of PULSE GAIN potentiometer and switch.

y. Verify elapsed time indicator operation.

z. Verify operation of standard serial interface.

aa. Verify SPL pushbutton switch operation.

bb. Verify ALT DIF pushbutton switch operation.

3.401.5 Control, Sensor C-9037()/AWG-9. The C&D TS shall provide the capability for performing the following tests on the sensor control assembly:

Verify proper operation of momentary double detent action switch (HALF/FULL).

b. Verify proper operation of MODE SELECT pushbutton switches.

c. Verify ELEV potentiometer vernier control operation.

d. Verify MRL momentary pushbutton switch operation.

e. Verify OFFSET momentary pushbutton switch operation.

f. Verify three position WCS toggle switch.

g. Verify three position IR/TV toggle switch.

h. Verify LIGHT TEST pushbutton switch operation.

i. Verify PWR RESET pushbutton switch operation.

j. Verify jewel light power indication operation.

(1) WCS

(2) IR/TV

(3) PWR RESET

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

k. Verify hand control X and Y positioning operation.

3.4.1.6 Display, Tactical IP-1110()/AWG-9. The C&D TS shall provide the capability for performing the following tests on the tactical information display (TID) assembly.

a. Using the appropriate test patterns, set up the displays and verify symbol generation strobes and numerics for:

(1) Presence

(2) Distortion

b. Display and verify flashing symbol capability.

c. Verify five position RANGE switch operation.

d. Verify DISPLAY control pushbutton switch operation.

e. Verify four position TID MODE rotary switch operation.

f. Verify BRIGHT potentiometer control operation.

g. Verify CONTRAST potentiometer control operation.

h. Verify STEERING mode drum indicator readouts.

i. Verify eight position DEST rotary switch operation.

j. Verify seven position NAV MODE rotary switch operation.

k. Verify selected numeric data readouts on the TID CRT.

l. Verify information drum indicator readouts.

m. Verify STBY AND READY indicator plaque operation.

n. Verify operation of standard aerial interface.

o. Verify elapsed time indicator operation.

p. Verify TRACK HOLD pushbutton switch operation.

q. Verify CLSN pushbutton switch operation.

3.4.1.7 Recorder, Mission Data KF-21 () and KF-22 (). The C&D TS shall provide the capability for performing the following tests on the mission data recorder:

a. Verify operation of mission data recorder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

(1) ON/OFF Camera motor power switch operation.

(2) Clock operation.

(3) Verify proper Film Low Indicate Signal.

(4) Displays test.

b. Verify elapsed time indicator operation.

3.5 Workmanship. Applicable.

3.6 C&D TS elements. Unless otherwise specified herein, elements of the C&D TS shall be installed or stored in rack-and-panel type cabinets.

3.6.1 Console, Test, Control-Display-Doppler Filter OJ 251()/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.1.1 Panel, Power Distribution SB-3695()/AWM-23(V). The power distribution panel shall provide the capability to:

a. Control the test station power function.

b. Indicate that input power forms are applied to the test station.

c. Provide overload protection for the test station.

d. Display test station malfunction indications.

3.6.1.2 Digital multimeter. A digital multimeter, Systron-Donner 9340-5B, or equivalent, shall be supplied and shall provide capability to perform ac and dc voltage, and resistance measurements.

3.6.1.3 Counter/timer. A counter/timer, Systron-Dormer 6038-EE-10A, or equivalent, shall be supplied and shall provide capability to perform freqency and time interval measurements.

3.6.1.4 Switching Unit, Fault Isolation SA-1894()/AWM-23(V). The fault isolation switching unit shall provide capability to perform the fault isolation switching during fault isolation tests of the WRAs and during test station self test.

3.6.1.5 Mounting, MT-4500()/AWM-23(V). The mounting shall serve as a holding fixture for Display, Detail Data IP-1111()/AWG-9, IP-1173()/AWG-9 and IP-1185()/AWG-9 and shall provide electrical connections for the WRA.

3.6.1.6 Mounting MT-4499()/AWM-23(V). The mounting shall serve as a holding fixture for Display, Tactical IP-1110()/AWG-9, Control, Sensor C-9037()/AWG-9, Control, Computer C-9036()/AWG-9, and Control, Sensor C-9035()/AWG-9 WRAs. The assembly shall also provide electrical and cooling air connections for the WRAs.

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T06:09Z
Check the source to verify that this is the currentversion before use.

3.6.1.7 Drawer, storage. The storage drawer (3.2.1.9.1) shall provide full panel width storage space for module extenders and special tools which are used in maintenance of the C&D TS and WRAs listed in Table IV.

3.6.1.7.1 Extender Kit, C&D TS. The extender kit shall provide the capability to extend modules from the respective WRAs or ground support equipment (GSE) assemblies. The kit shall be stored in the storage drawer of 3.6.1.7.

3.6.1.8 Adapter, test-hand control. The hand control test adapter shall serve as a holding fixture for Control, Sensor C-9037()/AWG-9 WRA.

3.6.2 Console, Test, Control-Display-Doppler Filter OJ-252()/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.2.1 Oscilloscope. An oscilloscope, Hewlett-Packard 181AR, with 1801A-001 and 1821A plug-ins, or equivalent, shall be supplied. The oscilloscope shall provide for display and observation of waveforms. The oscilloscope shall also provide for variable persistence of displayed signals, and for storage of signals.

3.6.2.2 Panel. Test. Electrical C-9178()/AWM-23(V). The electrical test panel shall provide the controls, displays and switching capability to conduct functional and fault isolation tests on the WRAs assigned to the C&D TS. Controls on the front panel shall be provided to command WRA Power On or Power Off, command emergency off for station power, select and enable functional or fault isolation tests, select and apply input signals to WRAs, select measurement instruments during test, command standard serial interface (SSI) word generator to provide the desired word, control setting of attenuators, and control level of simulated WRA inputs. Displays on the front panel shall be provided to display test number, test mode and status, and SSI BIT fault.

3.6.2.3 Panel , Coolant Supply MX-9306()/AWM-23(V). The coolant supply panel shall provide cooling air connections and cooling air supply for Filter-Multiplexer F-1331()/AWG-9 WRA. The panel shall also provide the electrical connectors and cables for Recorder, Mission Data KF-21() and KF-22() WRA.

3.6.2.4 Mounting, MT-4501()/AWM-23(V). The mounting shall provide mounting for Recorder, Mission Data KF-21() and KF-22() and Filter-Multiplexer F-1331()/AWG-9 WRAs. The mounting surface shall also provide the electrical connections to Filter-Multiplexer F-1331()/AWG-9 WRA.

3.6.2.5 Panel, blank. A blank panel shall cover the unused full width portion of bay no. 2.

3.6.2.6 Power Supply PP-6849()/AWM-23(V). The power supply shall provide the capability to supply dc voltages to WRAs and test station circuits, using another dc power supply (3.6.2.7) as a source. The power supply shall also provide the capability to sense the output power forms for overvoltage, undervoltage, and over-current for test station and WRA protection. The power supply shall also provide continuous monitoring capabilities of output power forms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

3.6.2.7 Power supply, dc. A dc power supply, Lambda LB-706-FM-3632, or equivalent, shall be' supplied and shall provide the capability to supply 200 Vdc at 10 amperes to the dc-dc converter panel of Power Supply PP-6849()/AWM-23(V).

3.6.3 Elapsed time meter. An elapsed time meter conforming to MIL-M-7793 shall be provided. The meter shall be readily visible during operation of the C&D TS and shall be capable of registering at least 9,999 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2,5 Acceptance test plan. Applicable. Table V shall apply.

4.2.6 Test results. Applicable.

4.2,7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test locations. Applicable.

4.6 Acceptance schedule. "Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:09Z
Check the source to verify that this is the current version before use.

TABLE V
ACCEPTANCE TEST PLAN FOR CONTROLS-DISPLAYS-
DOPPLER FILTER TEST STATION

OJ-214()/AWM-23(V)

1.0	TEST OBJECTIVE. To demonstrate self-test and operational run-in tests of the C&D test station.
2.0	TEST CONDITIONS
	a. AWG-9 Slave Units (none).
	b. Utilities Required (Information only, ref. 4.2.8)
	(1) Power as specified in 3.2.2.1.
	(2) Clean air cooling as specified by 3.2.2.2.
3.0	TESTS. In accordance with Acceptance Test Procedure 3282205.
	a. Perform self-test.
	b. Perform operational run-in tests.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The C&D TS is intended for intermediate level use to functionally test, fault isolate, and verify the serviceability of WRAs of the controls/displays and radar subsystems of Weapon Control System (WCS) AN/AWG-9.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross-reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

	Official Nomenclature	Contractor Part No.
a.	Filter-Multiplexer F-1331 ()/AWG-9	481042-1XX
b.	Control, Sensor C-9035 ()/AWG-9	481501-1XX
c.	Control, Computer C-9036 ()/AWG-9	481505-1XX
d.	Display, Detail Data IP-1111 ()/AWG-9, IP-1173()/AWG-9 and IP-1185()/AWG-9	481541-1XX
e.	Control, Sensor C-9037 ()/AWG-9	481560-1XX
f.	Display, Tactical IP-1110 ()/AWG-9	481580-1XX
g.	Recorder, Mission Data KF-21() and KF-22()	481590-1XX
h.	Test Station, Controls-Displays-Doppler Filter OJ-214 ()/AWM-23(V)	493250-1XX
i.	Console, Test, Control-Display-Doppler Filter OJ-251 ()/AWM-23(V)	493251-1XX
j.	Console, Test, Control-Display-Doppler Filter OJ-252 ()/AWM-23(V)	493252-1XX
k.	Kit, Tool, Special TK-222 ()/AWM-23(V)	493601-1XX

6.5 Liquid cooler. Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81783(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

AGC	Automatic gain control
C&DTS	Controls-displays-doppler filter test station
DDD	Detail data display
GSE	Ground support equipment
IF	Intermediate frequency
LO	Local oscillator
RF	Radio frequency
SSI	Standard serial interface
TID	Tactical information display

6.8 Convenience outlet power. Applicable.

6.9 Changes from previous issue are not marked.

Custodian:
Navy-AS

Preparing Activity:
Navy-AS

Project No. 4935-NO14

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22 R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

MIL-T-81783C(AS) Test Station, Controls-Displays-Doppler Filter OJ-214( )/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

☆ U S GOVERNMENT PRINTING OFFICE 1973—714 866/3983 2 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20361

POSTAGE AND FEES PAID

DOD 316



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20361

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:09Z
Check the source to verify that this is the current version before use.